Alexander Arias

Software Engineer | Software & Hardware Validation

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SUMMARY

Software Engineer with professional experience in Python-based hardware and software validation, debugging, and automated testing at Qualcomm. Background in systems programming, operating systems, computer architecture, object-oriented programming, and C, C++, and Java.

WORK EXPERIENCE

Qualcomm — San Diego, CA

Engineering Technician - Operations Department (Software & Hardware Validation)

June 2025 - May 2026

- Executed high-volume automated and manual validation workflows across 10,000+ devices using Agile methodologies and Python-based test systems in PyCharm spanning Snapdragon, PMIC, RF, digital, and related platforms.
- Troubleshot hardware and software failures, analyzed diagnostic results, reproduced defects, and documented debugging data using internal engineering and validation tools.
- Collaborated with engineers through iterative defect reproduction, debugging, regression testing, and fix verification cycles to maintain hardware/software validation reliability.
- Supported CI/CD pipeline testing and validation activities by executing automated Python-based tests, reproducing failures, verifying fixes, and documenting results across evolving hardware and software builds.
- Documented test results, validation metrics, failure behavior, and troubleshooting findings while consistently exceeding monthly performance targets.

University of California San Diego — San Diego, CA

Computer Science Outreach Volunteer - Eta Kappa Nu Engineering Honor Society

Jan. 2023 - Dec. 2023

- Delivered STEM outreach activities introducing fundamental computer science concepts to elementary and high school students.
- Supported interactive coding and algorithm demonstrations for student engagement in computing topics.
- Mentored high school students on academic pathways in computer science and engineering.

Computer Science Tutor - CSE Department

Sep. 2022 - Dec. 2022

- Tutored students in C and Assembly for Computer Organization and Systems Programming coursework.
- Provided one-on-one and group instruction to reinforce core concepts in low-level programming and computer architecture.

PROJECTS

Custom ISA and CPU *(SystemVerilog, Python, C, Git)*

- Led a 3-person software engineering team in system design and CPU architecture to implement a custom 9-bit instruction set architecture targeting Forward Error Correction workloads.
- Designed and implemented core CPU logic in SystemVerilog, including instruction decoding, control flow, and execution behavior.

Nachos Operating System *(Java, C, Linux, Git)*

- Contributed to a 4-person software engineering team implementing object-oriented Java operating-system components spanning thread synchronization, CPU scheduling, process management, and virtual memory using locks, semaphores, and condition variables.
- Implemented and debugged paging-based virtual memory and concurrent thread behavior, including page tables, page faults, synchronization issues, race conditions, and deadlocks.

Graphics Shadow Mapping and 3D Rendering Projects *(C++, C, OpenGL, Git)*

- Applied fragment shaders, Mandelbrot fractals, and 3D rotation algorithms; collaborated with one teammate on a software engineering project to develop a 3D OpenGL rendering application featuring dynamic shadow mapping.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, JavaScript, Assembly, SystemVerilog, HTML, CSS
Frameworks and Libraries: OpenGL, .NET, Node.js, JUnit
Software Engineering: Software Development Life Cycle (SDLC), Object-Oriented Programming & Design (OOP), Data Structures & Algorithms, Multithreading & Concurrency, Debugging, Software Testing, Unit Testing, Automated Testing
Development Tools: Git, GitHub, JetBrains IDEs (PyCharm, IntelliJ IDEA), Visual Studio, VS Code, GDB, Valgrind, Linux Command Line
Operating Systems: Linux, Windows

EDUCATION

University of California San Diego — San Diego, CA

Bachelor of Science in Computer Science

Dec. 2023